SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics COURSE CODE: CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5 Total Marks: 25

Code of Conduct
I R.Yazhini (192521190) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.Yazhini

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

CO2-AT3

APPLICATION BASED QUESTIONS

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Cloud auto-scaling automatically increases or decreases computing resources according to traffic demand. Load balancing distributes incoming requests across multiple servers to prevent overloading. Together they reduce downtime, improve response time, and provide a better user experience. Monitoring tools track CPU usage, traffic and response time to trigger scaling when needed.

For Example, an Ecommerce website like Amazon can add servers during festival sales such as diwali and reduce them when traffic falls

2. Cloud storage provides durability by storing multiple copies of data across different devices or locations. Redundancy protects backup data against hardware failures. Encryption protects data during storage and transmission, while access control ensures that only authorized users can access sensitive files. Cloud storage reduces hardware, maintenance, and electricity costs through pay-as-you-use pricing.

For Example, a company can restore its database from cloud backups after a fire or server failure at its main office.

3. A content Delivery Network (CDN) stores cached content on geographically distributed Edge servers. Users receive data from a near by server, reducing latency and improving loading speed. Edge servers also reduce the load on the main server and support smoother streaming and gaming experiences. Cloud providers offer

multiple regions and availability zones to improve global availability and fault tolerance

For Example, a global gaming application can use a CDN to deliver game updates and images quickly to players in different countries

4. cloud providers support compliance through security certifications, auditing tools, access controls and compliance reports. Logging records user activities, monitoring detects suspicious behaviors, and encryption protects financial data during storage and transmission. The institution should select cloud regions that satisfy applicable data sovereignty laws and configure storage and backups to remain within permitted locations

For Example, a bank can encrypt transaction records, restrict access to authorized employees, and maintain audit logs to support regulatory inspections

containers package an application with its dependencies and share the host OS kernel, making them lightweight and fast to start. VM run separate guest OS and generally consume more resources. Kubernetes automates container deployment, scaling, service discovery, and recovery when containers fail. containers are portable across compatible cloud environments, reducing vendor dependency

For Example, an online shopping application can run separate microservices for users login, product catalog, payments, and order in containers managed by Kubernetes